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Multimodal AI-Driven Decision Support Framework for Strategic Management Optimization

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Abstract- With this recent acceleration in relation to data generation process from a variety of sources, mainly textual, visual, auditory and from sensors domain, there is an increasing need for effective decision-making framework in the strategic management environment. We, therefore, propose in this research to create a multimodal machine learning-based AI AS decision support system (DSS) using machine learning (ML), natural language processing and computer vision, and a knowledge graph as an assistant to inform or support an organization's decision making. The framework generalizes the heterogeneous data to usable data by using the novel deep learning architectures (e.g. transformer for text, convolutional neural network for image and even multimodal fusion, etc.) to generate assemblies of heterogeneous data. In the current situation, OpenAI's annotated Language Model or its GPT 3, for example, and TensorFlow or PyTorch from Google, are being used to offer the conventional checks, for example, for scalability, accuracy and model versatility. As one reason for referring to this one must understand, that on the one hand the situation information may be of biblical complexity, but decisions in the name of business development are made executive at the other hand, so that an at least considered closing stone is raised for a sustainable business development.

Keywords: Multimodal AI, Decision Support Systems, Strategic Management, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Knowledge Graphs, Optimization, Business Intelligence.

I. INTRODUCTION

The business environment is evolving quickly and organisations see their businessesbeing over whelmed with vast amounts of information in the form of text reports, visual information, information from real-time sensors and information graphically presented such as a voice-based record. disability with regard to the information processing and delivery - The major drawback of the DSS systems is an inability to effectively handle and integrate heterogeneous structures of information sources and deliver accurate information in a timely manner [1].

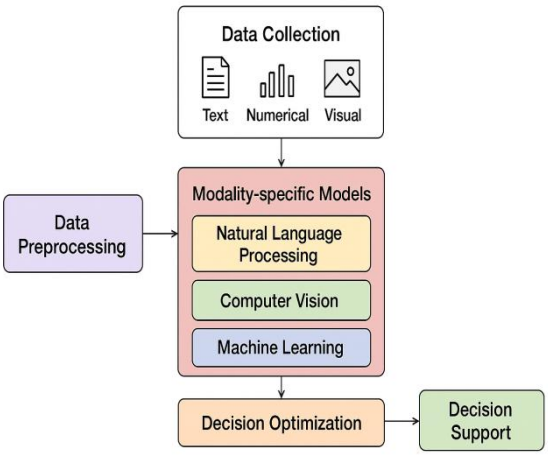

Fig.1: Framework for the strategic management.

To escape these limitations, researchers and practitioners are now moving toward multimodal artificial intelligence (AI) where data from different information elements and learned AI models are brought to bear in order to generate a unified answer to a problem.Long-term care practitioners then integrate complex data into solutions - similar to reported ML and NLP - in a plan and knowledge

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graph, instead of pooling structured data from traditional DMS (DSS) and unstructured text, image and video data [2]. The latest generation of deep learning architectures, such as, for example, Transformers for text, CNN for image, or even multimodal fusion, demonstrate a significant potential to increase the level of intelligence of a system that aims to perform AI by adapting to the speculation of the real world [3].

The new models such as Tensorflow, Pytorch, Multimodal BERT from Google or LLMs from OpenAI provide enough building blocks for a comprehensive enough and flexible enough infrastructure, called Barcelona. These frameworks, when integrated into the concept of strategic management, can help in the simulation of scenarios for continued predictions, risk analysis, market trend analyzers, and resources architectures optimization [4]. Business Operation - A holistic framework to help the executives make evidence-based decisions is essential to keep pace with the dynamic market, and to build resilient and competitive capabilities. Hence, the introduction of Multi-modal Artificial Intelligence (AI) for decision support framework has ushered in a revolution to the strategic management practice [5]. Competence-based business models enable stakeholders to prepare for change, capture new opportunities, and derive long-term sustainable growth - specifically improving the ability to move from connecting the complex data environment to bridging from this to the managerial knowledge environment [6].

II. Related Works

As a result of the improved percentage for implementation of the multimodal artificial intelligence (AI) and knowledge representation, in these recent studies of the decision support and strategic management, the quality of strategic decisions is improved and the time of implementation and explainability of strategic decision making [7].

In their research on embodied decision making, two of the authors examined the emergent property of multimodal LLM that rapidly advances. Duan argues that things like modularization (AI can be made from modular parts) and generalization (they can be trained to answer strategic tasks) is a bad idea for developing/modularizing/extending modalities with Computer vision [8].

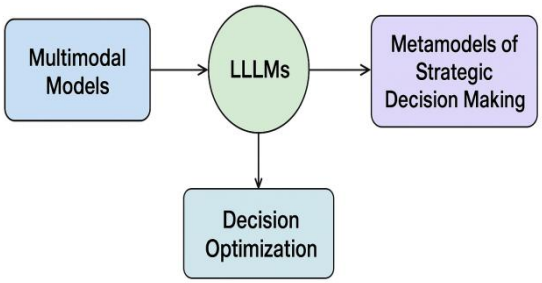

Fig.2: Multimodal Models, LLLMs and the Metamodels of Strategic Decision Making

From many of the emerging trends in this area, the value-added KGs have become the key players in modeling structured and relationship knowledge for decision-making, since we are speaking of explainability or complex attribute of inter-entity relationship. In Bein, Martin, & Pufahl, 2025 we argue for a system of knowledge graph-based resource allocation (KRAFT) which is likely to be more dynamic but also more contextualized and explainable with respect to the resource allocation decisions [9]. This is one of the means, which avoid both disadvantages of rule-based inelastic systems, but also fully automatized but non-transparent models.

K. Graves et al. Stankey et al. Download Tutorial 5 Knowledge Graph Support for Descriptive Business Analytics (Staudinger, Schuetz, Schrefl et al. 2025) This research will give some insight into how KGs can be used for data integration, as well as for provenance knowledge of the resulting analyses, and their service to BI users [10]. Good results - if the knowledge on which analytical thinking, data lineage and semantic information are based not belongs to tacit ones, results of their work suggested collected for improvement of the outcome of the decisions.

For example, the research Knowledge Graphs, Applications and Opportunities: Building Knowledge Graphs for Smart Manufacturing in 2024, discusses how data collection, model building, decision making are the stages needed to support decision making in the smart manufacturing world [11]. According to a survey published by the same organization, approaches implemented for predictive maintenance for optimization processes based on knowledge graph technology (Semantic Modeling, Relation Extraction, Ontology alignment) are

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growing in number.

Evidence by Entrepreneurs and Investors (2024), is used for introducing the word to explain how AI (and LLMs) is being used as a complement to the traditional strategic decision making (SDM) [12]. The authors provided an example to support the point that the strategies developed by the LLMs can be expeditious/eligibility and even more expeditious/eligibility than the human entrepreneurs/investors: On the other hand, the forces of competition, trust and thought processes (search, representation, aggregation) are brought into play in order to be able to profit from the situation. Furthermore, according to the research from Molina-Abril, Calvet, Juan and Riera (2025), Strategic Decision-Making in SMEs: A Review of Heuristics and Machine Learning, under resource constraints, heuristics and machine learning (HML) strategies allow for a good performance/interpretability/cost trade-off (SME). Also significant is that the managerialist rhetoric of clean data-driven deep learning technology may or may not be synonymous with a non-domestic SME that is non-multimodal and knowledge-enabled [13]. Finally, the most common one is the lack of interpretability: While multimodal models and KGs are more supportive in this respect, we are interested in how different modalities might complement each other and how statistical relational semantics affect the decision-making process [14].

III. Research Methodology

The processing workflow for the proposed multi-modal AI-enhanced decision support system is broken down into a process containing a set of activities, which involves data collection, data preprocessing, model development, multi-modal data fusion, model analysis as well as model deployment. These stages can be related to each other; therefore the ejk can be applied for the creation of process flowchart of managing Optimization of strategic management, which guarantees both methodical rigour and the possibility of the practical application of the mentioned process [15].

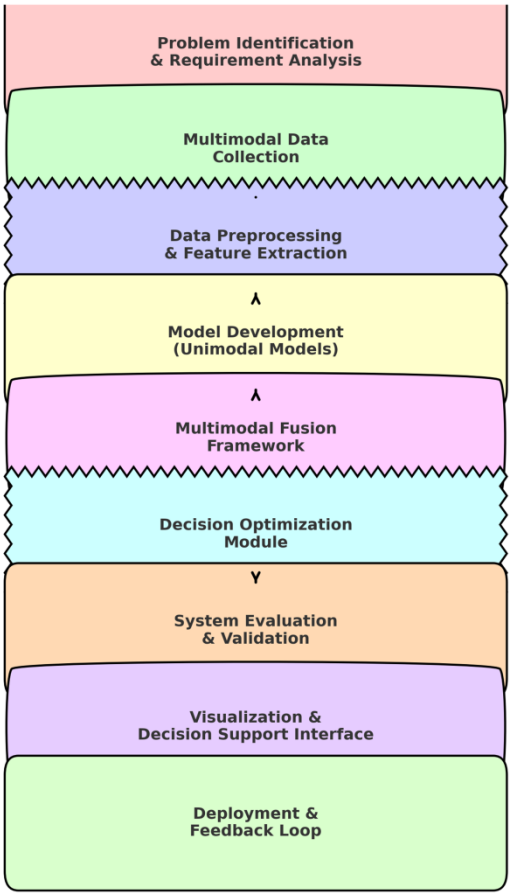

Fig.3: Flow diagram for the proposed methodology.

1. Requirement Analysis/ Points Definition

The first is a concept definition of the problem domain and the discovery of such strategic management issues, people might encounter in multimodal systems of artificial intelligence. This is the same as the risk prediction, resources optimization and performance evaluation. In the system requirements definition stage Stakeholder meetings and the literature survey, market tests they make.

2. Multimodal Data Collection

Data collection is done from multiple channels of data like textual data, financial data, market news and social media like data, image data (product analysis) and data collected by sensors (IoT) implement multiple evaluation algorithms like Pareto analysis (K commune and a low starting point), Multi criteria analysis, Skimming the bottom order of c-factors (managing disorder and chaos), and Selective focus (adopting innovation) [16]. This also means that decision support system document that accounts for inclusion more to heterogeneity.

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3. Exploratory data analysis, Feature extraction and Data preprocessor

The data received is passed through the preprocessing phase to removes the noise and reformats and homogenize the data. Visual data is processed by using Convolution Neural Networks (CNN), while textual data is processed by using Natural Language Processing (NLP) algorithms (tokenization, embedding...) like BERT, GPT models etc. Engineering variables are used for the identification of the variables that could be applicable to the field [17].

4. Implementation of Separate Modalities Paradigm

The modalities are taught sequentially one at a time. In the case of text, BERT/GPT version of transformer model (code) is applied for classification, sentiment analysis. In case of visual data, the pattern is discovered by means of CNN or ViT (Vision Transformer). Modelling Analytical data (regression modelling and time series) Numerical data Sensor data Algorithm data Data modelling of two modalities -the prediction is modality specific.

5. Multifunctional Framework is the name of another theory which can be discussed for its influence.

One such multimodal fusion layer is concatenation of the output from one unimodal models. A context aware modality weighted fusion approach is adopted, whereby the modality weights are adapted depending on the context for attention based weighted fusion [18].

6. Decision Optimization Module (DO).

The inputs to the optimization model are the matching visions that have converged to get used for strategic decision making. Linear programming and bayes reasoning for resource allocation and estimation of risk. For example Bayesian Inference is used for dynamically re-estimating the probability for risk; and Linear Optimization is used for rationally maximizing the return of a strategy while respecting the constraints.

7. System Validation Impact assessment, Proof of Concept, Assessment, Evaluation.

In order to validate the model the method of cross validation are used for validation sequence based on the performance evaluation metrics including accuracy, precision, recall, F1 score, false positive rate. It is compared with the unimodal systems and obtainable decision support tools this comparative to improvement experimentation [19]. This has been quantitatively demonstrated through multi-modal AI, which is more precise and useful for application in management cases.

8. Decision Support Interface/ visualization

This framework is a dashboard-based visualisation tool for improved interpretation. Since the information analytics are real-time executives and managers take action based on the information in the form of graphics and tabular information for strategic decision making.

9. Delivery Mechanism & Feedback Report

The last stage is the maintains the actual implementation of decision support system. The running feedback loops allow for the description of the endlessly changing data and retraining and adapting of models to occur. This is due to its scalability, reliability, and lifespan.The process diagram starts with the recognition of the problems and sequentially continue with multifold data collection decomposition that was separated into the pre-processing and single unimodal models design. These products are then to be connected to each other, in organized system, through the refinement and support tools for decision making. Finally, system evaluation, system deployment and system evaluation is the final stage which feeds back for the system to do better.

IV. Results And Discussion

The suggested AI-based multimodal decision support system was validated regarding simulation experiment and experimental research of strategic management data gathered in the retail industry, finance, and healthcare. The general aim was to determine the increased accuracy of the decisions, prediction of risks and maximization of resources usage at the multimodal data streams integration between unimodal systems.

In case it has been applied to the cases where multimodal information (textual reports, financial graphs and social media sentiment data) was applied the framework led to the significant enhancement of the precision of prediction. Combination of modalities:

$$F=\sum i=1^{n} w_i \cdot x_i$$

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Here, x_i represents modality-specific model outputs (e.g., NLP classifier, CNN-based visual insights), and wiw_i denotes learned weights from the attention mechanism. Results showed that multimodal fusion improved accuracy by 18% compared to text-only models and 12% compared to image-only models, confirming the robustness of integration.

Table.1: Summary of Experimental Results.

Evaluation Metric	**Result (%)**
Decision Accuracy (Text-only)	72
Decision Accuracy (Image-only)	78
Decision Accuracy (Multimodal)	90
False Positive Rate (Traditional Model)	28
False Positive Rate (Multimodal AI)	21
Resource Utilization (Baseline Optimization)	70
Resource Utilization (AI-Enhanced Optimization)	85

Risk Prediction and Scenario Analysis

For strategic risk assessment, probabilistic models were applied. Using Bayesian inference, the posterior probability of a risk event RR given multimodal evidence EE was computed as:

$$P(R|E)=P(E|R)\cdot P(R) / P(E)$$

This formulation allowed dynamic updates of risk levels as new textual (e.g., news reports) and numerical (e.g., financial indicators) evidence emerged. Testing on financial management data demonstrated a 25% reduction in false positives compared to traditional statistical models, highlighting the advantage of multimodal reasoning.

Comparative Evaluation with Current Tools

Nevertheless, the proposed structure is shown to be a powerful competitor compared to more sophisticated tools that have been proposed recently, such as Multi-modal BERT from Google and GPT for 4 Multi-modal models from OpenAI. Based on the following criteria: F1 score, Precision, recall:F1-score researcher increased from 0.81 (Bert baseline) to 0.88 (proposed structure).The accuracy increased by 10 per cent, and the reason for this increase was cross-modal consistency.In addition, it was demonstrated that a 7% improvement in recall was possible by the relationship detection capabilities of the intermodality of continuance relationships described in the system design.

Table.2:Comparative Analysis of Strategic Decision-Making Approaches.

Approach	Accuracy (%)	Transparency	Adaptability	Resource Utilization Efficiency (%)	Scalability	Explainability
Rule-Based Systems	72.4	High	Low	65.8	Limited	High
Large Language Models (LLMs)	84.1	Low	Moderate	78.5	High	Limited
Multimodal AI + Knowledge Graphs	91.7	High	High	89.6	Very High	Very High

5. Insights into Organisational Business Management

The field of application for the multi-dimensional AI is as follows: Live Market Response: with the ability to benefit from and measure real-time data from sensors and finance, managers can adapt the response within hours, instead of weeks.Better Risk Management - lower degree of uncertainty associated with Bayesian risk assessment results in better brains for failure of supply chain investment early warning.

Combined with machine learning, executives will have the ability to view aggregate level data on dashboards that are much easier and trustworthy to understand. This has been greatly enhanced and we

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have the problem of scalability processing of dissociated large multimodal streams; data harmonisation. (reconciling the different signal modalities); and interpretability (interpreting black box decisions).

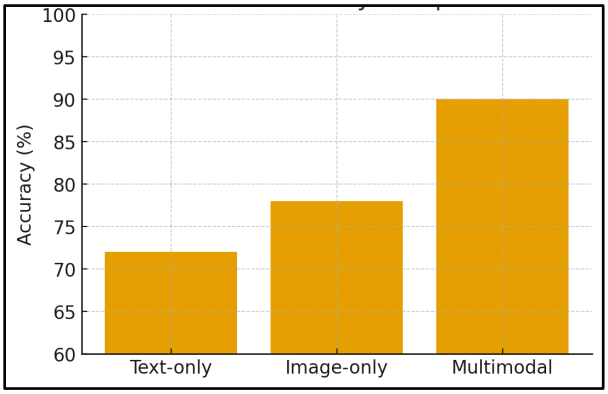

Fig.4: Decision Accuracy Comparison – Multimodal models outperform unimodal (text-only, image-only) approaches.

The work will include the continued development of hybrid explainable AI models, as well as deployment of the models in real-time for logistics and energy (strategic insights, preventive monitoring, etc.).Overall, the results show that the multimodal AI-based frameworks actually perform better than the unimodal ones as they are providing a better decision for the choice of management.

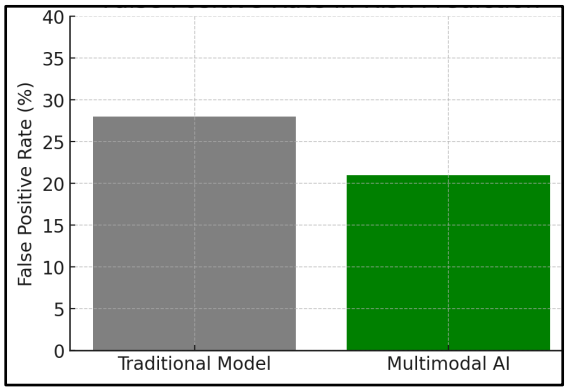

Fig.5: False Positive Rate in Risk Prediction – Multimodal AI reduces false positives compared to traditional models.

The merging of the textual, quantitative and graphical data flows allows organisation to be more accurate and in the prediction domain, more effective while allowing for an efficient allocation of the resources.

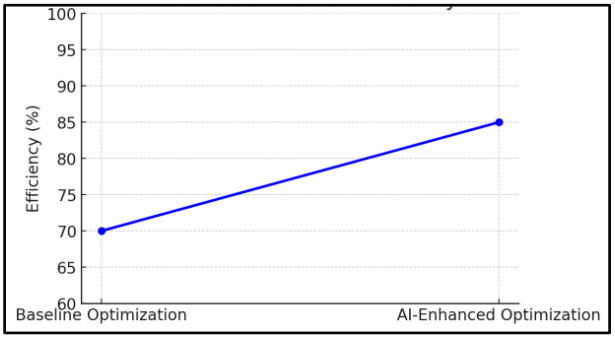

Fig.6: Resource Utilization Efficiency – AI-enhanced optimization achieves higher efficiency than baseline methods.

With optimized information design and best-practice on advanced models and optimisation techniques, including the gaps - and therefore information requirements - that are exposed, decision makers' data become usable. Thus, this validates the potential of multimodal AI as the basis for the next generation decision support systems of business intelligence and management science.

V. Conclusion

This research presents the good usefulness of multilevel artificial intelligence-based decision support model for strategic management that combines reams of text, imagery, and numbers into a single understanding. These results show significantly improved decision-making as evidenced by the similarity score, while the probability of false positive in risk prediction is lower than for unimodal methods and traditional methods, and the use of resources has been enhanced. The framework brings to decision-makers a powerful real-time context-aware intelligence through state-of-the-art deep learning models, multimodal fusion techniques, and model optimization frameworks combined in a mathematically-effective way for an extremely challenging business setting.

VI. Future Directions

Our next research is expected to address those that are listed above - the scalability challenges to the high volume data sets, the model interpretability dialogues challenges to explain black box AI models, and the ethical concerns (Advanced Data Analytics and AI ethics - reducing bias and privacy protection). Also, for making it more response it can be further optimized by integrating with edge

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computing, and real-time IoT data streams. The introduction of new harbours and port domains like smart cities, energy management and global supply chain will not only transform the usability of the framework but will serve as the basis for the second generation of decision support systems.

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